Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 05)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following water sources is usually the most convenient for large communities? Tick one box.

[1 mark]

A Desalinated seawater ☐

B River water ☐

C Rainwater catchment ☐

D Newly discovered aquifers ☐

0 1 . 2 Explain why the quality requirements for water differ between uses, using **two** named uses.

[2 marks]

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0 1 . 3 Explain why inter-basin transfers may be used and give **one** disadvantage.

[2 marks]

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0 2 . 1 Describe the process of anaerobic sludge digestion and give **two** benefits of the process.

[4 marks]

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0 2 . 2 Explain **two** problems of using sewage sludge as an agricultural fertiliser and how each can be reduced.

[4 marks]

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0 2 . 3 Explain why it is more efficient to have separate collection systems for foul sewage and for surface drainage water.

[2 marks]

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0 3 . 1 Which of the following properties would make a pesticide **least** likely to biomagnify? Tick one box.

[1 mark]

A High liposolubility and high persistence ☐

B Low liposolubility and low persistence ☐

C High liposolubility and low persistence ☐

D Low liposolubility and high persistence ☐

0 3 . 2 Explain the difference between **bioaccumulation** and **biomagnification**.

[2 marks]

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0 3 . 3 Explain why pesticides are described as an unusual type of pollutant.

[2 marks]

5

0 4 A coal-mine spoil heap on a hillside has been abandoned. The spoil contains shale with some iron pyrites and residual coal.

0 4 . 1 Explain **three** problems this spoil heap could cause and describe how each could be managed.

[6 marks]

0 4 . 2 Explain why deep layers of soil are applied when a spoil heap containing toxic materials is restored.

[2 marks]

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0 4 . 3 Explain why an old steel-works spoil heap may be valuable for wildlife conservation.

[2 marks]

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0 5 In the 1950s a chemical factory at Minamata, Japan, discharged inorganic mercury compounds into a shallow bay. Thousands of local people who ate fish and shellfish from the bay suffered severe neurological damage and many babies were born with birth abnormalities.

0 5 . 1 Explain how the low-toxicity inorganic mercury discharged into the bay came to cause such severe poisoning.

[4 marks]

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0 5 . 2 Explain why methyl mercury caused both neurological damage in adults and birth abnormalities.

[3 marks]

0 5 . 3 Compare the toxicity of the three chemical forms of mercury.

[3 marks]

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0 6 A group of students wants to investigate the effect of vegetation cover on rain-splash soil erosion.

0 6 . 1 Describe how they could carry out the investigation in the laboratory.

[5 marks]

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0 6 . 2 Give **three** variables, other than those you have standardised, that affect rain-splash erosion in the field.

[3 marks]

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0 6 . 3 Explain why it is easier to obtain valid results in the laboratory than in the field, and give **one** disadvantage of laboratory studies.

[2 marks]

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0 7 . 1 Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

A A waste is assumed to cause pollution until research shows it is unlikely to do so ☐

B The polluter must pay for any damage caused ☐

C Emissions should be as low as reasonably achievable ☐

D Only pollutants known to be toxic need to be controlled ☐

0 7 . 2 Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

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0 7 . 3 Give **two** choices about emission location that reduce the severity of pollution.

[2 marks]

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0 8 . 1 Explain the difference between a negative and a positive feedback mechanism.

[2 marks]

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0 8 . 2 Describe **two** negative feedback mechanisms that may reduce global warming.

[4 marks]

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0 8 . 3 Explain how melting permafrost and melting methane hydrate act as positive feedback mechanisms.

[4 marks]

0 8 . 4 Give **three** other positive feedback mechanisms.

[3 marks]

1

2

3

0 8 . 5 Explain what is meant by a climate **tipping point**.

[2 marks]

15

0 9 . 1 Explain how each of the following human activities changes the amount of carbon in the atmosphere.

[6 marks]

Ploughing farmland .....

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Draining marshland / peat bogs .....

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Toxic pollution of the sea .....

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0 9 . 2 Give **three** anaerobic environments created by human activities that release methane.

[3 marks]

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2 .....

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0 9 . 3 Explain how rising atmospheric CO_2 causes ocean acidification.

[1 mark]

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1 0 Table 1 shows the three forms of oxygen found in the stratosphere.

Table 1

Form of oxygen	Number of atoms per molecule
Monatomic oxygen (O)	1
Diatomic oxygen (O_2)	2
Triatomic oxygen / ozone (O_3)	3

1 0 . 1 Explain how the three forms of oxygen protect life at the surface from ultraviolet radiation.

[3 marks]

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1 0 . 2 Give **two** other ways in which the atmosphere supports life.

[2 marks]

1 .....

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2 .....

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1 0 . 3 Explain why the atmosphere is thermally stratified into layers.

[1 mark]

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1 0 . 4 The troposphere and stratosphere are the layers most affected by human activities. Discuss how human activities alter the energy processes in these two layers and evaluate the difficulties of predicting the consequences.

[9 marks]

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Write an essay on one of the following topics.

11 . 1 Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

11 . 2 Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS